

1. Two body gravitational acceleration

$$\underline{\underline{F}} = - \frac{G M m}{r^2} \frac{\underline{\underline{r}}}{r}$$

$$\mu = G M, \quad \hat{\underline{\underline{r}}} = \frac{\underline{\underline{r}}}{r}$$



$$F = ma$$

$$\therefore \underline{\underline{a}} = \frac{\underline{\underline{F}}}{m} = - \frac{G M}{r^2} \frac{\underline{\underline{r}}}{r}$$

$$\underline{\underline{a}} = - \frac{\mu}{r^3} \underline{\underline{r}}$$

$$\therefore \ddot{\underline{\underline{r}}} = - \frac{\mu}{r^3} \underline{\underline{r}}$$

2. Specific angular momentum.

$$\vec{h} = \vec{r} \times \vec{v}$$

시간에 따른 변화

$$\begin{aligned}\dot{\vec{h}} &= \dot{\vec{r}} \times \vec{v} + \vec{r} \times \dot{\vec{v}} \\ &= \cancel{\vec{v} \times \vec{v}} + \vec{r} \times \dot{\vec{v}} \\ &= \vec{r} \times \dot{\vec{v}}\end{aligned}$$

$$(\sin 0 = 0)$$

$$(\text{and, } \dot{\vec{v}} = \vec{a} = -\frac{\mu}{r^3} \vec{r})$$

$$\therefore \dot{\vec{h}} = \vec{r} \times \left(-\frac{\mu}{r^3} \vec{r} \right)$$

$$\boxed{\dot{\vec{h}} = 0}$$

$$\boxed{h = \text{constant.}}$$

↪ 크기면 $r^2 \dot{\phi}$.

3. Specific mechanical Energy.

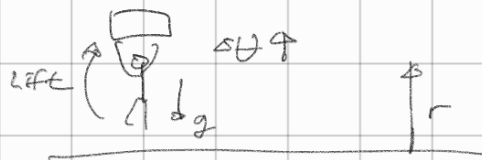
특정 기계적 에너지 = $\frac{T}{m} = \frac{v^2}{2} \quad \left(T = \frac{1}{2}mv^2 \right)$

중력 퍼텐셜 : $\frac{\theta}{m} = -\frac{GM}{r} = -\frac{\mu}{r}$

+) $F_r = -\frac{GMm}{r^2}$

$F_r = -\frac{d\theta}{dr}$ ← 중력의 크기 퍼텐셜 에너지 변화는 반대
 $0 \approx V_\infty = 0$

$\frac{d\theta}{dr} = \frac{GMm}{r^2}$



$\theta = \int \frac{GMm}{r^2} dr$

$= GMm \int r^{-2} dr$

$= -\frac{GMm}{r} + C$

$\theta_\infty = 0, \quad C = 0$

$\theta = -\frac{\mu m}{r} \quad \therefore \frac{\theta}{m} = -\frac{\mu}{r}$

$E = \frac{T}{m}$

$= \frac{T}{m} + \frac{\theta}{m}$

$$E = \frac{v^2}{2} - \frac{\mu}{r}$$

→ if $v \downarrow, r \downarrow \quad E < 0$

궤도 에너지가 부족

if $E > 0$

궤도 에너지가 충분

